

Test I
CS602/CO434 Image Processing
September, 2025

Marks: 10

Time: 30 minutes

1. Provide mathematical and graphical representation of image negative. 1
2. If you apply powerlaw transformation with an exponent less 1, what will be nature of the output image? Justify your answer. 1
3. Differentiate between sampling and quantization. 1
4. Consider the following image for answering the questions given below.

4	2	5	2
2	5	2	5
6	4	6	2
4	2	4	4

- a) Find the dynamic range of the image. 1
- b) Perform range transformation so that the image acquire new dynamic range 0-15. 2
- c) Find the histogram of the given image. 1
- d) Perform histogram sliding of the given image with a factor of 2. 1
- e) Perform histogram equalization of the image. 2

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